For this assignment use the dataset provided in Class Activity-2. Each student must select one Indian state from the below list

['West Bengal', 'Delhi', 'Haryana', 'Jharkhand', 'Goa', 'Karnataka', 'Kerala', 'Madhya Pradesh', 'Maharashtra', 'Tamilnadu', 'Chhattishgarh', 'Telengana', 'Andhra Pradesh', 'Himachal Pradesh', 'Punjab', 'Rajasthan', 'Gujarat', 'Uttarakhand', 'Uttar Pradesh', 'Sikkim', 'Assam', 'Arunachal Pradesh', 'Nagaland', 'Manipur', 'Mizoram', 'Tripura', 'Meghalaya' , 'Bihar', 'Jammu and Kashmir', 'Odisha']

Use the **provided code to extract data** for the selected state. (Copy the given code and use, don't modify it in the shared notebook). The **state name must exactly match** the spelling and capitalization used in the list.

After selecting the state and extracting the data for the state, answer the questions given below.

Submit **two files**:

- One PDF with all figures and analysis
- One Jupyter Notebook (.ipynb) with complete code

File naming format (mandatory): Name_roll-no_A1.extension

Q.1 Analyze how temperature and precipitation have changed over your chosen state during 1951–2022, and compare these changes with the all-India mean.

In your analysis, address the following:

- What are the long-term trends in mean temperature and precipitation over the state during 1951–2022? **5**
- Is the warming rate over the state higher or lower than the all-India average? **2**
- Between daytime maximum temperature (Tmax) and nighttime minimum temperature (Tmin), which is increasing faster over your state? **3**
  How does this compare with the all-India behaviour? **2**
- Does precipitation show a clear monotonic trend similar to temperature, or does it behave differently? Explain with proper reasoning. **5**

Q.2 Examine how the seasonal characteristics of temperature and precipitation have evolved over time. Divide the period 1951–2022 into three sub-periods (early, mid, and recent decades as per your choice) and investigate:

- How does the seasonal cycle of temperature and precipitation changes across these periods? **9**
- How summer-season maximum temperatures and winter-season minimum temperatures have evolved over time? **5**
- Whether there are noticeable changes in the timing, intensity, or duration of the monsoon rainfall season? You may comment based on changes in monthly rainfall distribution, seasonal concentration, or shifts in peak rainfall. **6**

Q.3 Compare the statistical distribution of temperature and precipitation over selected state and all-India mean for the same time period. Use appropriate visualizations (e.g., histograms, PDFs, boxplots) to compare distributions.

Using this comparison, discuss:

- How does the mean climate of the state differ from the all-India mean? **5**
- Whether extremes are more pronounced over the state w.r.t all-India mean? **3**
- What these differences reveal about the spatial heterogeneity of the Indian climate system? **2**

Q.4 Investigate how climate extremes over your chosen state have changed during 1951–2022, and compare these changes with India-wide behaviour. The definitions of the extreme indices **TXx**, **TNx**, and **R95pTOT** are provided in the *Class Activity–2 slides (PDF)* already shared with you, and **must be followed exactly**.

Your analysis should include:

- Calculate the climate extreme indices (TXx, TNx, R95pTOT) for your own state and show the evolution in extreme from 1951-2022. **6+9**
- How do trends in extremes differ from trends in the corresponding mean variables? **9**
- How does the state-level evolution of extremes compare with the all-India mean changes? **6**